OffSec PEN-200
12 Week Learning Plan

Welcome to OffSec PEN-200! We are delighted to offer a customized learning plan designed to support your learning journey and ultimately enhance your preparedness for the Offensive Security Certified Professional (OSCP) certification.

The Learning Plan comprises a week-by-week journey, which includes a recommended studying approach, estimated learning hours, course topics to focus on, topic exercises, capstone exercises, and challenge machines to complete, as well as supplemental materials to reinforce your learning (if you so choose).

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Week 1

Overview and Study Approach	The first four topics serve as an introduction to the course material and provide a general approach to the course. We also recommend learners explore the following OffSec communication mediums: • OffSec Help and FAQs: https://help.offsec.com/hc/en-us • Discord: How may I join the OffSec Community? • Youtube Channel: @OffSecTraining • Twitch Channel: @offsecofficial • Twitter: @OffSecTraining
Learning Topics	1) Penetration Testing with Kali Linux: General Course Information 2) Introduction To Cybersecurity 3) Effective Learning Strategies 4) Report Writing for Penetration Testers 5) Information Gathering 6) Vulnerability Scanning
Exercises	6.2.1. Whois Enumeration 6.2.2. Google Hacking 6.2.3. Netcraft 6.2.4. Open-Source Code 6.3.1. DNS Enumeration 6.3.2. TCP/UDP Port Scanning Theory 6.3.3. Port Scanning with Nmap 6.3.4. SMB Enumeration 6.3.5. SMTP Enumeration 6.3.6. SNMP Enumeration 7.1.1. How Vulnerability Scanners Work 7.1.2. Types of Vulnerability Scans 7.1.3. Things to consider in a Vulnerability Scan 7.2.1. Installing Nessus 7.2.2. Nessus Components 7.2.3. Performing a Vulnerability Scan 7.2.4. Analyzing the Results 7.2.5. Performing an Authenticated Vulnerability Scan 7.2.6. Working with Nessus Plugins 7.3.1. NSE Vulnerability Scripts 7.3.2. Working with NSE Scripts
Estimate Time (Hours)	20
Supplemental Learning*	**Videos:** • OffSec Academy Recordings: OSA - PEN - 200: Week 1A - Active Info. Gathering • Offsec Live Twitch Recordings: Get to Know the Minds Behind PEN-200 (2023) **Relevant Labs:** N/A

Week 2

Overview and Study Approach	This week, we will focus on the basic methodology, techniques, and tools required to perform successful enumeration and exploitation of web and common web application attacks.
Learning Topics	1) Introduction to Web Application Attacks 2) Common Web Application Attacks
Exercises	8.2.4. Security Testing with Burp Suite 8.3.3. Enumerating and Abusing APIs 8.4.5. Privilege Escalation via XSS 9.1.1. Absolute vs Relative Paths 9.1.2. Identifying and Exploiting Directory Traversals 9.1.3. Encoding Special Characters 9.2.1. Local File Inclusion (LFI) 9.2.2. PHP Wrappers 9.2.3. Remote File Inclusion (RFI) 9.3.1. Using Executable Files 9.3.2. Using Non-Executable Files 9.4.1. OS Command Injection
Estimate Time (Hours)	24
Supplemental Learning*	**Videos**: • OffSec Academy Recordings: OSA - PEN - 200: Week 2A - Web Application Attacks • OffSec Live Twitch Recordings: WEB-200: Cross-site Scripting **Relevant Labs**: • Proving Ground Practice labs: ◦ Key examples: helpdesk, hetemit ◦ Additional practice (when ready): Apex, xposedapi, reconstruction, slort, payday, uc404

Week 3

Overview and Study Approach	The learners will focus on SQL Injection which is one of the most common web application vulnerabilities. Additionally, learners will acquire the skills to conduct target reconnaissance, explore exploitation scenarios using malicious Microsoft Office documents and Windows Library files.
Learning Topics	1) SQL Injection Attacks 2) Client-side Attacks
Exercises	10.1.2. DB Types and Characteristics 10.2.3. Blind SQL Injections 10.3.2. Automating the Attack 11.1.1. Information Gathering 11.1.2. Client Fingerprinting 11.2.1. Preparing the Attack 11.2.2. Installing Microsoft Office 11.2.3. Leveraging Microsoft Word Macros 11.3.1. Obtaining Code Execution via Windows Library Files
Estimate Time (Hours)	20
Supplemental Learning*	**Videos**: • OffSec Academy Recordings: OSA - PEN - 200: Week 3B - Web Application Attacks **Relevant Labs:** • Proving Ground Practice labs: ◦ Key examples: butch, Hepet ◦ Additional practice (when ready): hawat, pebbles, megavolt

Week 4

Overview and Study Approach	Focus on online resources that provide public known vulnerabilities exploits. Additionally, we will also examine offline tools within Kali that contain local-hosted exploits and learn techniques for overcoming any potential obstacles when utilizing these tools.
Learning Topics	1) Locating Public Exploits 2) Fixing Exploits
Exercises	12.1.1. A Word of Caution 12.2.1. The Exploit Database 12.3.1. Exploit Frameworks 12.3.2. SearchSploit 12.3.3. Nmap NSE Scripts 12.4.1. Putting It Together 13.1.3. Cross-Compiling Exploit Code 13.1.4. Fixing the Exploit 13.1.5. Changing the Overflow Buffer 13.2.2. Selecting the Vulnerability and Fixing the Code 13.2.3. Troubleshooting the "index out of range" Error
Estimate Time (Hours)	20
Supplemental Learning*	**Videos:** N/A **Relevant Labs:** • Proving Ground Practice labs: ◦ Key examples: Kevin, shifty ◦ Additional practice (when ready): Twiggy

Week 5

Overview and Study Approach	We will cover multiple techniques for detecting malicious software, as well as exploring methods to bypass AV software on target machines. Learners will also delve into network attacks, password cracking, and attacks against Windows-based authentication implementations.
Learning Topics	1) Antivirus Evasion 2) Password Attacks
Exercises	14.1.3. Detection Methods 14.2.2. In-Memory Evasion 14.3. AV Evasion in Practice 14.3.2. Evading AV with Thread Injection 14.3.3. Automating the Process 15.1.1. SSH and RDP 15.1.2. HTTP POST Login Form 15.2.1. Introduction to Encryption, Hashes and Cracking 15.2.2. Mutating Wordlists 15.2.3. Cracking Methodology 15.2.4. Password Manager 15.2.5. SSH Private Key Passphrase 15.3.1. Cracking NTLM 15.3.2. Passing NTLM 15.3.3. Cracking Net-NTLMv2 15.3.4. Relaying Net-NTLMv2
Estimate Time (Hours)	20
Supplemental Learning*	**Videos:** • OffSec Academy Recordings: OSA - PEN - 200: Week 7A • OffSec Live Twitch Recordings: PEN-200 (2023): Antivirus Evasion **Relevant Labs:** • Proving Ground Practice labs: ◦ Key examples: Authby, nickel ◦ Additional practice (when ready): Phobos

Week 6

Overview and Study Approach	Once we gain access to the target machine, we will need to escalate the privileges in order to perform more advanced actions on the compromised system. These topics will focus on techniques and exploits that enable successful privilege escalation on both Windows and Linux systems.
Learning Topics	1) Windows Privilege Escalation 2) Linux Privilege Escalation
Exercises	16.1.1. Understanding Windows Privileges and Access Control Mechanisms 16.1.2. Situational Awareness 16.1.3. Hidden in Plain View 16.1.4. Information Goldmine PowerShell 16.1.5. Automated Enumeration 16.2.1. Service Binary Hijacking 16.2.2. Service DLL Hijacking 16.2.3. Unquoted Service Paths 16.3.1. Scheduled Tasks 16.3.2. Using Exploits 17.1.2. Manual Enumeration 17.1.3. Automated Enumeration 17.2.1. Inspecting User Trails 17.2.2. Inspecting Service Footprints 17.3.1. Abusing Cron Jobs 17.3.2. Abusing Password Authentication 17.4.1. Abusing Setuid Binaries and Capabilities 17.4.2. Abusing Sudo 17.4.3. Exploiting Kernel Vulnerabilities
Estimate Time (Hours)	24
Supplemental Learning*	**Videos:** • OffSec Academy Recordings: ◦ OSA - PEN - 200: Week 4A - Privilege Escalation ◦ OSA - PEN - 200: Week 5A - Privilege Escalation ◦ OSA - PEN - 200: Week 6A Privilege Escalation **Relevant Labs:** • Proving Ground Practice labs: ◦ Key examples: morbo, Nibbles ◦ Additional practice (when ready): billyboss, escape

Week 7

Overview and Study Approach	We will cover port redirection and tunneling techniques using SSH. The topic will begin with simple techniques and gradually progress to more complex ones as we move towards more secure network environments.
Learning Topics	1) Port Redirection and SSH Tunneling
Exercises	18.2.3. Port Forwarding with Socat 18.3.1. SSH Local Port Forwarding 18.3.2. SSH Dynamic Port Forwarding 18.3.3. SSH Remote Port Forwarding 18.3.4. SSH Remote Dynamic Port Forwarding 18.3.5. Using sshuttle 18.4.1. ssh.exe 18.4.2. Plink 18.4.3. Netsh
Estimate Time (Hours)	20
Supplemental Learning*	**Videos:** • OffSec Academy Recordings: ◦ OSA - PEN - 200: Week 8A - Port Redirection and tunneling ◦ OSA - PEN - 200: Week 8B - Port Redirection and Tunneling **Relevant Labs:** • Proving Ground Practice labs: ◦ Key examples: N/A ◦ Additional practice (when ready): Nukem

Week 8

Overview and Study Approach	There may be many restrictions implemented on a network. We will focus on learning and leveraging various tunneling tools and strategies to bypass technologies such as deep packet inspection. We will also cover the Metasploit Framework, including its features, usage and its internal workings. By doing this, we can understand how these frameworks can assist us in real penetration tests.
Learning Topics	1) Tunneling Through Deep Packet Inspection 2) The Metasploit Framework
Exercises	19.1.2. HTTP Tunneling with Chisel 19.2.1. DNS Tunneling Fundamentals 19.2.2. DNS Tunneling with dnscat2 20.1.1. Setup and Work with MSF 20.1.2. Auxiliary Modules 20.1.3. Exploit Modules 20.2.1. Staged vs Non-Staged Payloads 20.2.2. Meterpreter Payload 20.2.3. Executable Payloads 20.3.1. Core Meterpreter Post-Exploitation Features 20.3.2. Post-Exploitation Modules 20.3.3. Pivoting with Metasploit 20.4.1. Resource Scripts
Estimate Time (Hours)	20
Supplemental Learning*	**Videos:** • OffSec Academy Recordings: OSA - PEN - 200: Week 10A - The Metasploit Framework **Relevant Labs:** • Proving Ground Practice labs: ◦ Key examples: Splodge, Internal ◦ Additional practice (when ready): wombo

Week 9

Overview and Study Approach	Focus on Active Directory (AD) enumeration, AD explore authentication mechanisms and learn where Windows caches authentication objects such as password hashes and tickets, after that, we'll get familiar with the attack methods targeting these authentication mechanisms.
Learning Topics	1) Active Directory Introduction and Enumeration 2) Attacking Active Directory Authentication
Exercises	21.2.1. Active Directory - Enumeration Using Legacy Windows Tools 21.2.2. Enumerating Active Directory using PowerShell and .NET Classes 21.2.3. Adding Search Functionality to our Script 21.2.4. AD Enumeration with PowerView 21.3.1. Enumerating Operating Systems 21.3.2. Getting an Overview - Permissions and Logged on Users 21.3.3. Enumeration Through Service Principal Names 21.3.4. Enumerating Object Permissions 21.3.5. Enumerating Domain Shares 21.4.1. Collecting Data with SharpHound 21.4.2. Analysing Data using BloodHound 22.1.1. NTLM Authentication 22.1.2. Kerberos Authentication 22.1.3. Cached AD Credentials 22.2.1. Password Attacks 22.2.2. AS-REP Roasting 22.2.3. Kerberoasting 22.2.4. Silver Tickets 22.2.5. Domain Controller Synchronization
Estimate Time (Hours)	16
Supplemental Learning*	**Videos:** • **OffSec Academy Recordings:** ◦ OSA - PEN - 200: Week 11A - Active Directory Attacks (Part 1) ◦ OSA - PEN - 200: Week 12A - Active Directory Attack (Part 1) ◦ OSA - PEN - 200: Week 13A - Active Directory Attack (Part 2) ◦ OSA - PEN - 200: Week 14A - Active Directory Attack (Part 2) • **OffSec Live Twitch Recordings:** ◦ PEN-200 (2023): Active Directory Enumeration ◦ Walkthrough of a PEN-200 AD Set **Relevant Labs: N/A** ◦

Week 10

Overview and Study Approach	Explore different lateral movement techniques that allow us to authenticate to a system and gain code execution using a user's hash or a Kerberos ticket.
Learning Topics	1) Lateral Movement in Active Directory
Exercises	23.1.1. WMI and WinRM 23.1.2. PsExec 23.1.3. Pass the Hash 23.1.4. Overpass the Hash 23.1.5. Pass the Ticket 23.1.6. DCOM 23.2.1. Golden Ticket 23.2.2. Shadow Copies
Estimate Time (Hours)	20
Supplemental Learning*	**Videos:** • **OffSec Academy Recordings:** ◦ OSA - PEN - 200: Week 11A - Active Directory Attacks (Part 1) ◦ OSA - PEN - 200: Week 12A - Active Directory Attack (Part 1) ◦ OSA - PEN - 200: Week 13A - Active Directory Attack (Part 2) ◦ OSA - PEN - 200: Week 14A - Active Directory Attack (Part 2) • **OffSec Live Twitch Recordings:** ◦ PEN-200 (2023): Active Directory Enumeration ◦ Walkthrough of a PEN-200 AD Set **Relevant Labs: N/A**

Week 11

Overview and Study Approach	The final topic will cover a complete penetration testing scenario. The remaining time will be devoted to organizing and consolidating all the notes taken on learning concepts from previous weeks, as well as completing exercises.
Learning Topics	1) Assembling the Pieces
Exercises	N/A
Estimate Time (Hours)	20
Supplemental Learning*	N/A

Week 12

Overview and Study Approach	The aim is to simulate an exam environment and assess your preparedness while identifying any areas that may require further attention. The time should be utilized to attempt to complete any of the OSCP grade labs (OSCP A, OSCP B, or OSCP C) in under 24 hours. These are retired OSCP exams.
Learning Topics	N/A
Exercises	N/A
Supplemental Learning*	N/A
Estimate Time (Hours)	20